

Intravascular Imaging- vs Angiography-Guided Complex PCI

5-Year Outcomes From a Randomized Trial

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ABSTRACT

BACKGROUND Data regarding long-term outcomes of intravascular imaging-guided percutaneous coronary intervention (PCI) for complex coronary artery lesions, compared with angiography-guided PCI, remain limited.

OBJECTIVES The authors sought to present long-term clinical outcomes of intravascular imaging-guided PCI compared with angiography-guided PCI in patients with complex coronary artery lesions.

METHODS Eligible patients with complex coronary artery lesions were randomly assigned 2:1 to undergo intravascular imaging-guided PCI or angiography-guided PCI in this prospective multicenter open-label superiority trial performed in South Korea. The primary endpoint was a composite of cardiac death, target vessel-related myocardial infarction, or clinically driven target vessel revascularization.

RESULTS A total of 1,639 patients underwent randomization with 1,092 assigned to imaging-guided PCI and 547 assigned to angiography-guided PCI. At a median follow-up of 5.3 years (Q1-Q3: 4.4-6.2 years), the primary endpoint occurred in 109 of 1,092 patients (10.5%) in the intravascular imaging-guided PCI group and 78 of 547 patients (14.9%) in the angiography-guided PCI group (HR: 0.68; 95% CI: 0.51-0.91; $P = 0.009$). Cardiac death or target vessel-related myocardial infarction occurred in 78 patients (7.6%) in the intravascular imaging-guided PCI group and in 56 patients (10.7%) in the angiography-guided PCI group, clinically driven target vessel revascularization in 45 (4.4%) and 32 (6.2%), and definite stent thrombosis in 1 (0.1%) and 4 (0.7%), respectively. There were no apparent differences in procedure-related safety events between the groups.

CONCLUSIONS In patients with complex coronary artery lesions undergoing PCI, intravascular imaging guidance reduced the risk of a composite of cardiac death, target vessel-related myocardial infarction, or clinically driven target vessel revascularization at median follow-up of 5.3 years, compared to angiography guidance. (Intravascular Imaging- Versus Angiography-Guided Percutaneous Coronary Intervention for Complex Coronary Artery Disease [RENOvATE-COMPLEX-PCI]; [NCT03381872](https://clinicaltrials.gov/ct2/show/study/NCT03381872)) (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS AND ACRONYMS

IVUS = intravascular
ultrasound

OCT = optical coherence
tomography

PCI = percutaneous coronary
intervention

Coronary angiography has been a standard method to guide percutaneous coronary intervention (PCI) for patients with coronary artery disease. Although the use of second-generation drug-eluting stents improved clinical outcomes after PCI, patients with complex coronary artery lesions still have higher risk of adverse clinical events after angiography-guided PCI than patients without complex coronary artery lesions. Intravascular imaging can provide additional information about lesion characteristics; appropriate size and length of stents; and acute procedural results including expansion, edge dissection, or apposition of the stents that can increase the short- and long-term risks of stent failure.¹⁻³ In recent years, multiple randomized controlled trials and meta-analyses have shown that PCI guided by intravascular imaging such as intravascular ultrasound (IVUS) and optical coherence tomography (OCT) reduces the risk of adverse clinical events compared with angiography-guided PCI.⁴⁻¹⁰ Based on accumulated evidence from recent trials, the American College of Cardiology/American Heart Association/Society for Cardiovascular Angiography and Interventions guideline and European Society of Cardiology/European Association for Cardio-Thoracic Surgery guidelines have upgraded the use of IVUS or OCT to a Class I recommendation for optimizing PCI in anatomically complex lesions.^{11,12} Nevertheless, recent trials of intravascular imaging-guided PCI reported clinical outcomes within 2 years from randomization,⁴⁻⁸ and evidence for sustained benefits of intravascular imaging-guided PCI beyond 2 years is lacking.

The RENOVATE-COMPLEX-PCI (Randomized Controlled Trial of Intravascular Imaging Guidance versus Angiography-Guidance on Clinical Outcomes After Complex Percutaneous Coronary Intervention) was the first trial to demonstrate prognostic benefit of intravascular imaging-guided PCI especially for reduction of hard clinical endpoints in all-comers

with complex coronary artery disease. Here, we describe the 5-year follow-up of the RENOVATE COMPLEX-PCI, which is the longest follow-up report, to investigate whether intravascular imaging-guided PCI would improve long-term clinical outcomes compared with angiography-guided PCI in patients with diverse complex coronary artery lesions.

METHODS

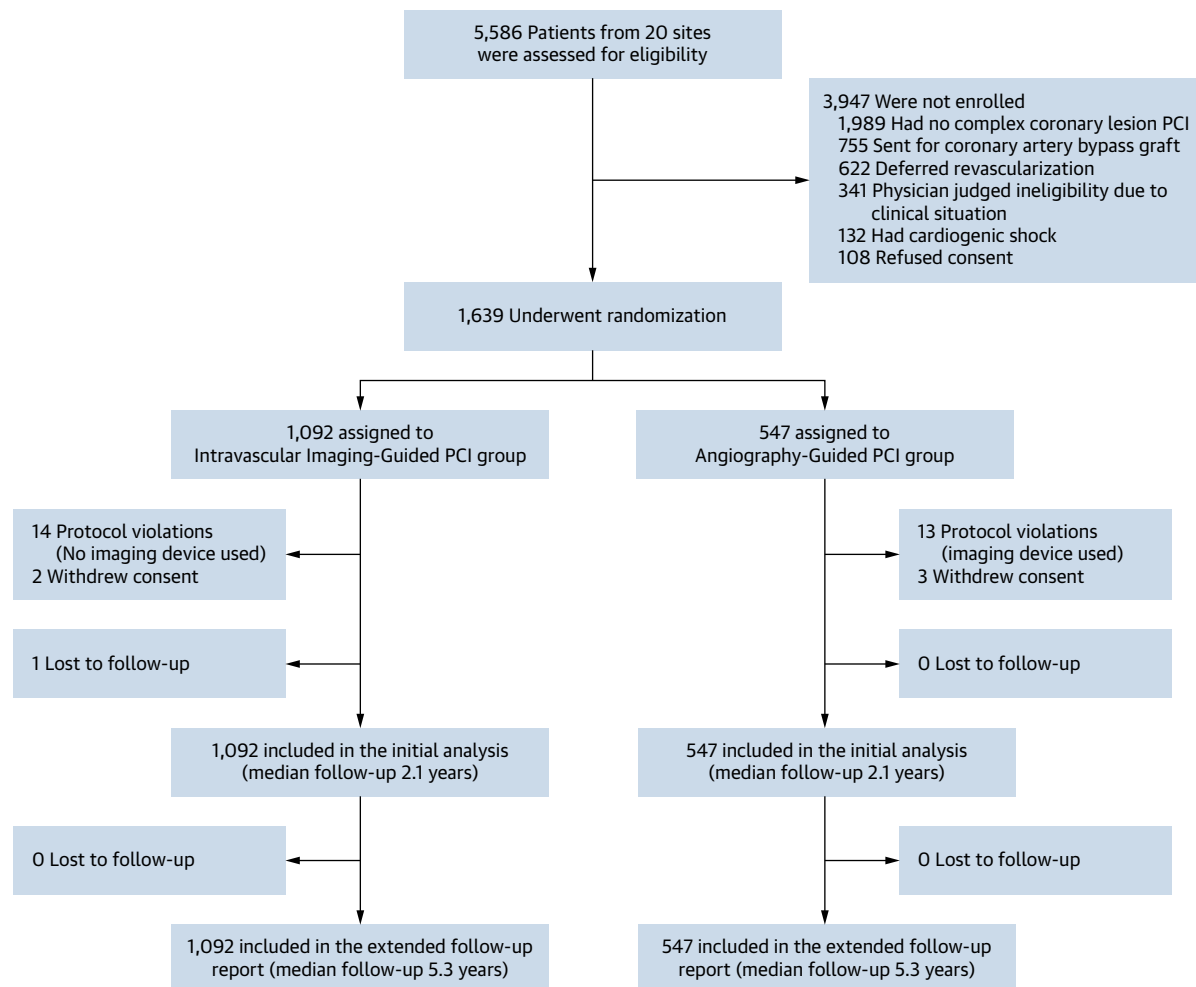
TRIAL DESIGN AND OVERSIGHT. The design and primary results of the RENOVATE COMPLEX-PCI have been reported previously.⁴ In brief, this investigator-initiated, prospective, randomized, open-label, multicenter trial enrolled patients with complex coronary artery disease from 20 hospitals in South Korea. Patients were randomized in a 2:1 ratio to intravascular imaging-guided PCI or angiography-guided PCI using an internet-based allocation system, stratified by clinical presentation (stable ischemic heart disease vs acute coronary syndrome) and by participating centers. All patients provided written informed consent before randomization.

EXTENDED FOLLOW-UP AFTER PRIMARY REPORT. Following publication of the primary results based on a median 2.1 years of follow-up, the study protocol was amended to extend annual surveillance until 4 years after the last patient enrollment. Trial protocol was revised with incorporation of plans for the extended follow-up period and was approved by the Institutional Review Board at each participating site. Scheduled in-person assessments were performed annually after the first year, and when clinic visits were not feasible, structured telephone interviews were conducted by trained coordinators. To ensure completeness of follow-up, survival status and major adverse cardiovascular events were cross-validated against the Korean National Health Insurance Service database, which provides nationwide coverage. A Data and Safety Monitoring Board oversaw the trial, and an independent Clinical Event Adjudication Committee whose members were masked to study

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

Manuscript received October 3, 2025; revised manuscript received December 23, 2025, accepted January 7, 2026.

FIGURE 1 Randomization, Treatment, and Follow-Up of Patients

Between May 2018 and May 2021, a total of 1639 patients with complex coronary artery lesions with plans for PCI underwent randomization in a 2:1 manner to intravascular imaging-guided PCI or angiography-guided PCI. A total of 14 patients in the intravascular imaging-guided PCI group did not have intravascular imaging devices used during the PCI procedure and 13 patients in the angiography-guided group had an intravascular imaging device used during the PCI. Two patients in the intravascular imaging-guided PCI group and 3 patients in the angiography-guided PCI group withdrew their consent. One patient in the intravascular imaging-guided PCI group was lost to follow-up. Following publication of the primary results based on a median 2.1 years of follow-up, the study protocol was amended to extend annual surveillance until 4 years after the last patient enrollment. No patient was lost to follow-up during the extended follow-up period. All patients were analyzed as an intention-to-treat population. PCI = percutaneous coronary intervention.

group assignments assessed all clinical events including additional events after the primary report (see the [Supplemental Appendix](#)). The Executive Committee and all the authors vouch for the accuracy and completeness of the data and for the fidelity of the trial to the protocol. The updated trial protocol and statistical analysis plan are available online.

PATIENTS. Eligible patients were ≥ 19 years of age with complex coronary artery lesions who planned to

undergo PCI. Complex coronary artery lesions were defined as true bifurcation lesions according to the Medina classification system with a side branch diameter of ≥ 2.5 mm, a chronic total occlusion, unprotected left main coronary artery disease, long coronary artery lesions with an expected stent length of ≥ 38 mm, multivessel PCI involving ≥ 2 major epicardial coronary arteries that are treated at the same time, a lesion requiring multiple stents (≥ 3 planned stents), in-stent restenosis, severely

TABLE 1 Characteristics at Baseline of the Patients Who Underwent Randomization

	Total (N = 1,639)	Imaging-Guided PCI (n = 1,092)	Angiography-Guided PCI (n = 547)
Demographic and clinical characteristics			
Age, y	65.6 ± 10.2	65.3 ± 10.3	66.0 ± 10.0
Male	1,300 (79.3)	869 (79.6)	431 (78.8)
Initial presentation			
Stable ischemic heart disease	807 (49.2)	532 (48.7)	275 (50.3)
Acute coronary syndrome	832 (50.8)	560 (51.3)	272 (49.7)
Unstable angina	534 (32.6)	361 (33.1)	173 (31.6)
Acute myocardial infarction	298 (18.2)	199 (18.2)	99 (18.1)
Non-ST-segment elevation myocardial infarction	258 (15.7)	171 (15.7)	87 (15.9)
ST-segment elevation myocardial infarction	40 (2.4)	28 (2.6)	12 (2.2)
Medical history			
Hypertension	1,005 (61.3)	682 (62.5)	323 (59.0)
Diabetes mellitus	617 (37.6)	394 (36.1)	223 (40.8)
Insulin-treated diabetes mellitus	51 (3.1)	28 (2.6)	23 (4.2)
Dyslipidemia	840 (51.3)	560 (51.3)	280 (51.2)
Current smoking	307 (18.7)	212 (19.4)	95 (17.4)
Chronic renal insufficiency	296 (18.1)	203 (18.6)	93 (17.0)
Previous PCI	395 (24.1)	268 (24.5)	127 (23.2)
Previous myocardial infarction	117 (7.1)	75 (6.9)	42 (7.7)
Previous stroke	112 (6.8)	70 (6.4)	42 (7.7)
Peripheral arterial disease	44 (2.7)	27 (2.5)	17 (3.1)
LV ejection fraction (%)	58.7 ± 11.6	58.4 ± 11.9	59.3 ± 11.0
Discharge medication			
Aspirin	1,606 (98.0)	1,069 (97.9)	537 (98.2)
P2Y ₁₂ inhibitor			
Any	1,603 (97.8)	1,067 (97.7)	536 (98.0)
Clopidogrel	1,217 (74.3)	799 (73.2)	418 (76.4)
Ticagrelor	209 (12.8)	148 (13.6)	61 (11.2)
Prasugrel	178 (10.9)	120 (11.0)	58 (10.6)
Oral anticoagulant	75 (4.6)	46 (4.2)	29 (5.3)
Statin	1567 (95.6)	1041 (95.3)	526 (96.2)
Beta blocker	710 (43.3)	466 (42.7)	244 (44.6)
ACE inhibitor or ARB	945 (57.7)	622 (57.0)	323 (59.0)
Target lesion characteristics			
Complex coronary lesions			
True bifurcation lesion	359 (21.9)	233 (21.3)	126 (23.0)
Chronic total occlusion	319 (19.5)	220 (20.1)	99 (18.1)
Unprotected left main coronary artery disease	192 (11.7)	138 (12.6)	54 (9.9)
Long coronary lesion	898 (54.8)	617 (56.5)	281 (51.4)
Multivessel PCI	622 (37.9)	409 (37.5)	213 (38.9)
Multiple stents (at least 3)	305 (18.6)	208 (19.0)	97 (17.7)
In-stent restenosis	236 (14.4)	158 (14.5)	78 (14.3)
Severely calcified	231 (14.1)	157 (14.4)	74 (13.5)
Ostial disease	251 (15.3)	182 (16.7)	69 (12.6)
Number of complex coronary lesions ≥ 3	505 (30.8)	352 (32.2)	153 (28.0)
Number of vessels with disease			
1-vessel disease	526 (32.1)	342 (31.3)	184 (33.6)
2-vessel disease	621 (37.9)	420 (38.5)	201 (36.7)
3-vessel disease	492 (30.0)	330 (30.2)	162 (29.6)

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calcified lesions, or ostial lesions of a major epicardial coronary artery. Major exclusion criteria were patients with coronary lesions that were not amenable for PCI as determined by the operator; cardiogenic

shock (Killip class IV) at presentation; known hypersensitivity or contraindication to aspirin, clopidogrel, prasugrel, ticagrelor, heparin, everolimus, or contrast media; or who were pregnant or breast

TABLE 1 Continued

	Total (N = 1,639)	Imaging-Guided PCI (n = 1,092)	Angiography-Guided PCI (n = 547)
Procedural characteristics			
Total no. of target lesions treated	1.5 ± 0.7	1.5 ± 0.7	1.5 ± 0.7
Intravascular imaging devices used ^a	1,091/1,639 (66.6)	1,078/1,092 (98.7)	13/547 (2.4)
Intravascular ultrasound	813/1,091 (74.5)	800/1,078 (74.2)	13/13 (100.0)
Optical coherence tomography	278/1,091 (25.5)	278/1,078 (25.8)	0/13 (0.0)
Volume of contrast media used, ml	207.3 ± 116.5	214.2 ± 118.5	193.7 ± 111.3
Procedural time, min	65.0 (47.0-89.0)	70.0 (51.0-95.0)	53.5 (40.0-75.0)
Procedural success	1,613 (98.4)	1,073 (98.3)	540 (98.7)

Values are mean ± SD, n (%), or median (IQR). ^a14 patients in the intravascular imaging-guided PCI group did not have intravascular imaging used during their PCI procedure due to failure to pass the device (n = 9), failed PCI (n = 4), or hemodynamic instability during procedure (n = 1). 13 patients in the angiography-guided group had intravascular imaging used during the procedure at the operator's discretion for treatment of long coronary lesion (n = 2), unprotected left main disease (n = 2), chronic total occlusion (n = 2), severe calcification (n = 3), unclear lesion length (n = 2), ostial lesion (n = 1), or hemodynamic instability during procedure (n = 1).

ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; HDL = high-density lipoprotein; LD = low-density lipoprotein; LV = left ventricle; PCI = percutaneous coronary intervention.

feeding. Full definition of complex coronary artery lesions and detailed inclusion and exclusion criteria are included in the [Supplemental Appendix](#).

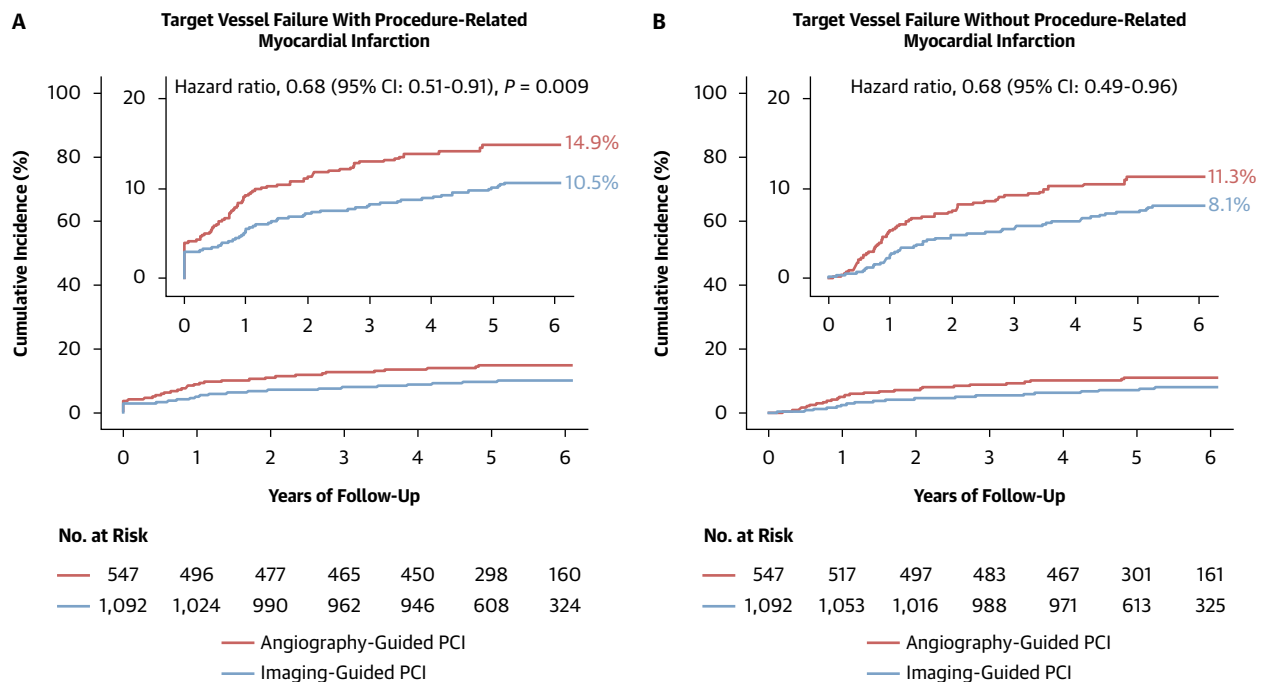
TREATMENT. Procedural techniques and optimization criteria for intravascular imaging-guided and angiography-guided PCI were reported in the primary publication and are summarized in the [Supplemental Appendix](#).¹ Briefly, all patients were treated with second-generation drug-eluting stents (everolimus-eluting stents with biodegradable or biocompatible polymer). In the imaging-guided arm, IVUS or OCT was used at the operators' discretion, and poststent optimization was mandatory based on predefined criteria for minimal stent area, apposition, and absence of edge dissection ([Supplemental Appendix](#)).¹ Additional postdilatation or stenting was recommended if optimization was not achieved, with repeat imaging to confirm final results. In the angiography-guided arm, optimization was based on visual angiographic criteria of the angiographic residual diameter stenosis was <10% of the reference vessel diameter without flow limiting coronary artery dissection. All angiograms and intravascular imaging data were analyzed by independent core laboratories after completion of the PCI procedure ([Supplemental Appendix](#)). Adjunctive medical therapy followed current clinical guidelines, with dual antiplatelet therapy recommended for at least 3-6 months for stable disease and 6-12 months for acute coronary syndrome, with adjustments at physician discretion.

TRIAL ENDPOINTS. The primary endpoint was target vessel failure, which was a composite of cardiac death, target vessel-related myocardial infarction, or clinically driven target vessel revascularization.

Secondary endpoints included individual components of the primary endpoint, target vessel failure without procedure-related myocardial infarction, a composite of cardiac death or target vessel-related myocardial infarction, definite stent thrombosis, contrast-induced nephropathy, and bleeding according to the definitions from the Bleeding Academic Research Consortium and TIMI.⁴ Safety endpoints, such as periprocedural complications, were adjudicated as described previously. Detailed definitions of clinical events are described in the [Supplemental Appendix](#).

STATISTICAL ANALYSIS. The trial was powered to determine the superiority of intravascular imaging-guided PCI over angiography-guided PCI with respect to the primary endpoint through follow-up until 1 year from the last patient enrollment. Details of the initial sample size assumptions are provided in the [Supplemental Appendix](#).

Analyses were conducted according to the intention-to-treat principle. For the primary endpoint, the cumulative incidence of target vessel failure was estimated, and treatment groups were compared using the Fine-Gray method to account for noncardiac death as a competing risk.¹³ As a sensitivity analysis, time-to-event analyses were also carried out using Kaplan-Meier estimates, with effect sizes expressed as HRs and 95% CIs derived from Cox proportional hazards regression. The proportional hazards assumption was evaluated with Schoenfeld residuals, tested at a 2-sided significance level of 0.05. Prespecified subgroup analyses of the primary endpoint were undertaken according to the type of intravascular imaging modality and selected clinical

FIGURE 2 Cumulative Incidence of Study Outcomes

(A) The primary end point was target vessel failure, which is a composite of cardiac death, target vessel-related myocardial infarction, or clinically driven target vessel revascularization in the intention-to-treat population during the overall study period (i.e., beginning from the time of randomization to the day of the first occurrence of a primary end point event, the day of the last office or phone visit, or the day of death during follow-up). The primary end point analysis included an estimation of the cumulative risk of target vessel failure and a comparison of randomization groups with method of Fine and Gray to adjust for the potential competing risk of non-cardiac death. Therefore, the hazard ratio indicates sub-hazard ratios from competing risk analysis. (B) Target vessel failure without procedure-related myocardial infarction is compared between the 2 groups. MI = myocardial infarction; PCI = percutaneous coronary intervention.

characteristics. Missing baseline data were not imputed. All models were adjusted for the patients' clinical presentation and participating centers (stratification factors in the randomization). Statistical analyses were performed using R version 4.1.2 (R Foundation for Statistical Computing) and IBM SPSS Statistics for Windows version 20.0 (IBM Corp).

RESULTS

A total of 1,639 patients with complex coronary artery lesions underwent randomization between May 28, 2018 and May 28, 2021 with 1,092 patients assigned to intravascular imaging-guided PCI and 547 assigned to angiography-guided PCI (Figure 1). Baseline clinical and procedural characteristics were well balanced between the groups (Table 1). Among the study population, 807 of 1,639 patients (49.2%) presented with stable ischemic heart disease and 832 of 1,639 patients (50.8%) presented with an acute coronary syndrome. Target lesion characteristics

showed that the most prevalent complex lesion characteristics were long coronary lesion, which was followed by multivessel PCI, true bifurcation lesion, and chronic total occlusion. Of the 1,639 patients, 505 (30.8%) had ≥ 3 characteristics of complex coronary artery lesions in the target lesion. Both groups showed similar baseline lesion severity assessed by the minimum lumen diameter, percentage diameter stenosis, and lesion length (Supplemental Table 1). Among patients assigned to the intravascular imaging-guided PCI group, intravascular imaging could not be used in 14 patients; IVUS was used in 800 of 1,092 patients (73.3%); and OCT was used in 278 of 1,092 patients (25.5%).

PROCEDURAL CHARACTERISTICS. In the intravascular imaging-guided PCI group, 1180 of 1569 target lesions (75.2%) were evaluated by intravascular imaging in both pre- and post-PCI phases. The use of intravascular imaging resulted in more frequent use of adjunctive balloon dilation of the stent with non-compliant balloons (Supplemental Table 1). In the

TABLE 2 Primary and Secondary Endpoints by Competing Risk Analysis^a

	Total (N = 1,639)	Imaging-Guided PCI (n = 1,092)	Angiography-Guided PCI (n = 547)	Sub-HR (95% CI)	P Value
Primary endpoint					
Target vessel failure ^b	187 (12.0)	109 (10.5)	78 (14.9)	0.68 (0.51-0.91)	0.009
Secondary end points^c					
Target vessel failure without procedure-related MI	140 (9.1)	82 (8.1)	58 (11.3)	0.68 (0.49-0.96)	0.027
Cardiac death or target-vessel related MI	134 (8.6)	78 (7.6)	56 (10.7)	0.68 (0.48-0.96)	0.029
All-cause death	133 (8.6)	86 (8.4)	47 (9.1)	0.89 (0.62-1.27)	0.516
Cardiac death	65 (4.4)	37 (3.8)	28 (5.6)	0.65 (0.40-1.07)	0.090
Myocardial infarction	86 (5.3)	51 (4.8)	35 (6.5)	0.78 (0.50-1.23)	0.290
Target-vessel related MI	75 (4.6)	43 (4.0)	32 (5.9)	0.72 (0.44-1.17)	0.190
Spontaneous MI ^d	24 (1.5)	13 (1.2)	11 (2.1)	0.75 (0.31-1.83)	0.530
Procedure-related MI ^e	52 (3.2)	30 (2.7)	22 (4.0)	0.71 (0.40-1.27)	0.250
Nontarget vessel related MI	12 (0.8)	8 (0.8)	4 (0.8)	1.34 (0.36-5.03)	0.670
Repeat revascularization ^f	121 (7.7)	73 (7.0)	48 (9.2)	0.80 (0.55-1.16)	0.240
Target vessel revascularization	77 (5.0)	45 (4.4)	32 (6.2)	0.73 (0.46-1.16)	0.180
Target lesion revascularization	60 (3.8)	35 (3.4)	25 (4.8)	0.74 (0.43-1.25)	0.250
Nontarget vessel revascularization	57 (3.7)	34 (3.2)	23 (4.5)	0.78 (0.45-1.32)	0.350
Definite stent thrombosis ^g	5 (0.3)	1 (0.1)	4 (0.7)	0.17 (0.02-0.90)	0.036
Contrast induced nephropathy ^h	40 (2.4)	26 (2.4)	14 (2.6)	0.98 (0.49-1.95)	0.950
Bleeding, nontraumatic spontaneous					
BARC type 2, 3, or 5	51 (3.3)	35 (3.5)	16 (3.0)	1.00 (0.52-1.95)	0.999
BARC type 3 or 5	29 (1.9)	22 (2.2)	7 (1.3)	1.50 (0.55-4.13)	0.430
TIMI major	15 (1.0)	14 (1.4)	1 (0.2)	5.50 (0.71-42.70)	0.100

Values are n (%) unless otherwise indicated. ^aFor clinical outcomes, including cardiac death, noncardiac death was treated as a competing event; For other outcomes, all-cause death was treated as a competing event. The database for the analysis was locked on May 10, 2025. Clinical end points were evaluated in the intention-to-treat population during the overall study period (i.e., beginning from time of randomization to the day of the first occurrence of a primary end-point event, the day of the last office or phone visit, or the day of death during follow-up). Percentages are the cumulative incidence at 6 years. All the models were adjusted for the clinical presentation and participating centers (stratification factors). ^bPrimary end point is target vessel failure, which is a composite of cardiac death, target vessel MI, target vessel revascularization. ^cThe individual end points listed are the first occurrence of that event. ^dSpontaneous myocardial infarction is defined using the 3rd universal definition of myocardial infarction. ^eProcedure-related myocardial infarction is defined as the Society for Cardiovascular Angiography and Interventions (SCAI) definition. ^fRepeat revascularization includes all first clinically indicated elective, urgent, or emergent revascularization procedures that were not planned during the index hospitalization during the overall study period. ^gDefinite stent thrombosis is defined as Academic Research Consortium criteria. ^hContrast-induced nephropathy is defined as an increase in serum creatinine of ≥ 0.5 mg/dL or $\geq 25\%$ from baseline within 48-72 hours after contrast agent exposure. Event rate is presented as proportion among groups.

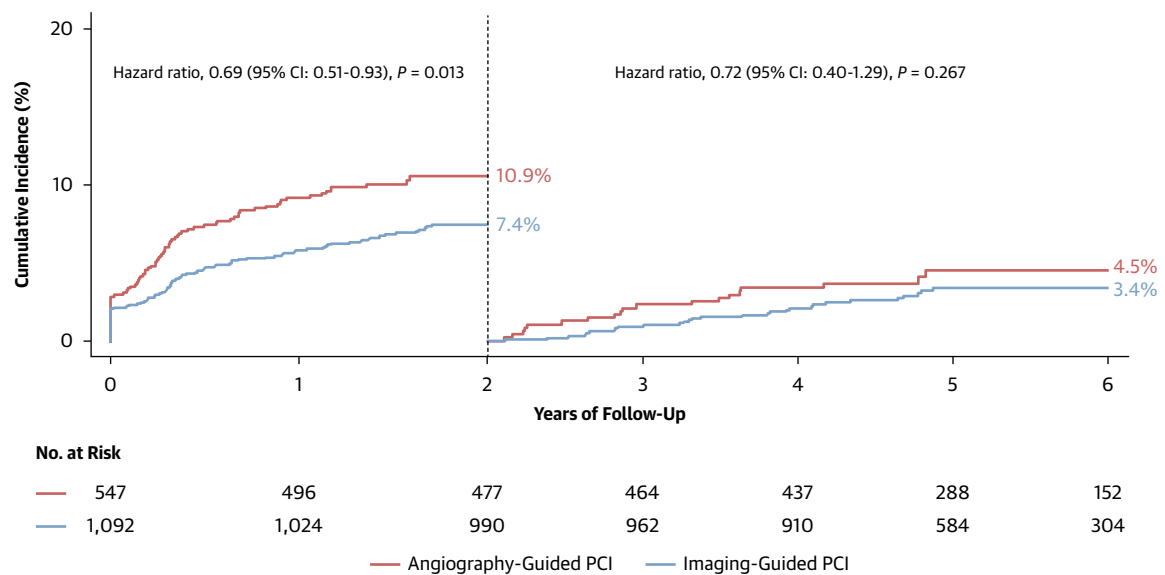
BARC = Bleeding Academic Research Consortium; MI = myocardial infarction; PCI = percutaneous coronary intervention.

intravascular imaging-guided PCI group, stent optimization was achieved in 496 of 1092 patients (45.4%) with stent optimization achieved in 339 of 800 patients (42.4%) imaged with IVUS and 157 of 278 patients (56.5%) imaged by OCT, respectively. Among lesions evaluated by IVUS, 659 of 1188 lesions (55.5%) met all stent optimization criteria while among lesions evaluated by OCT, 238 of 361 lesions (65.9%) met all stent optimization criteria ([Supplemental Table 2](#)). In the angiography-guided PCI group, angiographic stent optimization was achieved in 322 of 547 patients (58.9%). Procedural time was about 15 minutes longer in the intravascular imaging-guided PCI group than angiography-guided PCI group. Both groups showed similar rates of procedure-related complications during the index hospitalization ([Supplemental Table 3](#)).

STUDY ENDPOINTS. Since publication of the primary report on May 4, 2023, all enrolled patients completed follow-up until 4 years after the last patient enrollment and database was locked on

May 10, 2025. At median follow-up of 5.3 years (IQR: 4.4 to 6.2 years), the primary end point occurred in 109 of 1092 patients (10.5%) in the intravascular imaging-guided PCI group and 78 of 547 patients (14.9%) in the angiography-guided PCI group (hazard ratio, 0.68; 95% confidence interval, 0.51 to 0.91; $P=0.009$) ([Figure 2](#), [Table 2](#)). The estimated restricted mean survival time at median follow-up duration (5.3 years) was 59.8 months (95% CI: 58.8-60.7) with intravascular imaging-guided PCI and 57.4 months (95% CI: 55.9-58.9) with angiography-guided PCI (difference, 2.4 months; 95% CI: 0.6-4.1; $P=0.010$). The risk of target vessel failure without procedure-related myocardial infarction appeared lower in the intravascular imaging-guided PCI group than the angiography-guided PCI group (8.1% vs. 11.3%; HR: 0.68; 95% CI: 0.49-0.96; $P=0.027$) ([Figure 2](#)).

[Table 2](#) presents individual components of the primary and secondary end points. The incidence of all-cause death was 8.4% in the intravascular

FIGURE 3 Landmark Analysis At 2 Years After Initially Reported Time Point

A landmark analysis of the primary end point at 2 years from index procedure. The early period was beginning from the time of randomization to the day of the first occurrence of a primary end point event, the day of the last office or phone visit, or the day of death during follow-up before 2 years from index procedure. The late period was 2 years after index procedure until 4 years after the last patient enrollment. Since the initial publication of the trial was the results at median follow-up of 2.1 years, we selected 2 years as landmark point.

imaging-guided PCI group and 9.1% in the angiography-guided PCI group (HR: 0.89; 95% CI: 0.62-1.27; $P = 0.516$). [Supplemental Table 4](#) summarizes presumed cause of cardiac death, time until cardiac death, and relationship to target vessel. The incidence of cardiac death or target-vessel related myocardial infarction was 7.6% in the intravascular imaging-guided PCI group and 10.7% in the angiography-guided PCI group (HR: 0.68; 95% CI: 0.48-0.96; $P = 0.029$). Clinically driven target vessel revascularization occurred in 45 (4.4%) and 32 (6.2%) patients, respectively (HR: 0.73; 95% CI: 0.46-1.16; $P = 0.180$). Although there was no significant difference between the 2 groups regarding the risk of target lesion revascularization, the rates of edge restenosis were numerically lower in the intravascular imaging-guided PCI group than the angiography-guided PCI group (rates of edge restenosis: 8 of 35 target lesion revascularization in the intravascular imaging-guided PCI group [22.9%] vs. 10 of 25 target lesion revascularization in the angiography-guided PCI group [40.0%]). During extended follow-up period, there was no additional event of definite stent thrombosis, and the incidence was 0.3% in the overall study population and was lower in the intravascular imaging-guided group than in the

angiography-guided group (0.1% vs. 0.7%; HR: 0.17; 95% CI: 0.02-0.90; $P = 0.036$).

A landmark analysis at 2 years from index procedure found that the prognostic benefit of intravascular imaging-guided PCI was primarily observed during the earlier period (7.4% vs. 10.9%; HR: 0.69; 95% CI: 0.51-0.93; $P = 0.013$), while event rates beyond 2 years were numerically lower but not statistically significant (3.4% vs. 4.5%; HR 0.72; 95% CI: 0.40-1.29; $P = 0.267$) ([Figure 3](#), [Table 3](#)). When hazard ratios were plotted continuously over time, the treatment effect remained directionally favorable beyond 2 years, and the early benefit appeared to be maintained without late catch-up ([Supplemental Figure 1](#)).

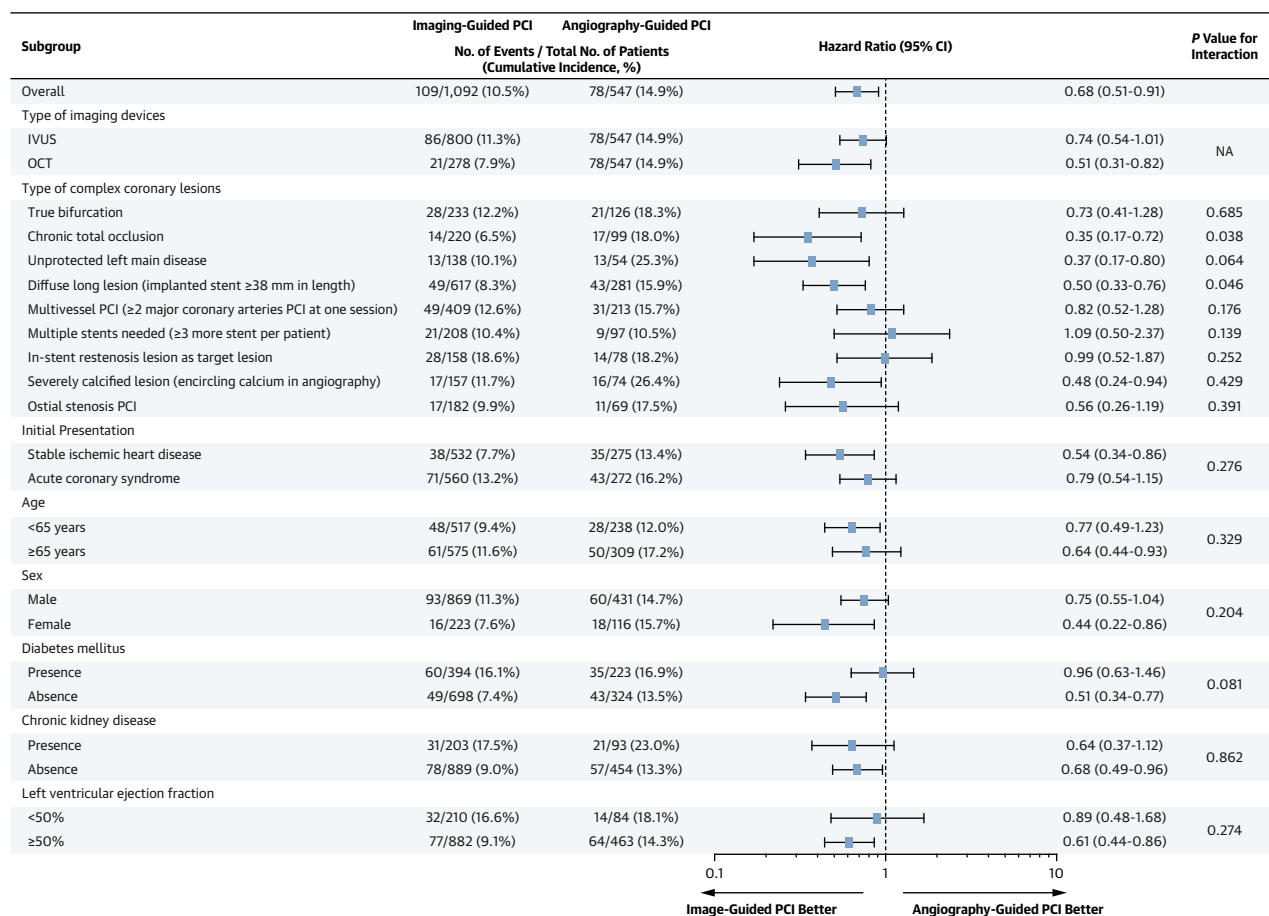
Prespecified subgroup analysis found consistent benefits of intravascular imaging-guided PCI than angiography-guided PCI across various subgroups ([Figure 4](#)). Notably, within the spectrum of complex coronary artery lesion subsets, the benefits of intravascular imaging-guided PCI over angiography-guided PCI were most apparent in chronic total occlusion and diffuse long lesion. Although there was no significant interaction between clinical presentation and treatment effect of intravascular imaging, patients with stable ischemic heart disease showed

TABLE 3 Landmark Analysis at 2 Years for the Primary and Secondary Endpoints

	Total	Imaging-Guided PCI (n = 1,092)	Angiography-Guided PCI (n = 547)	HR (95% CI)	P Value
Baseline to 2-years					
Primary end point					
Target vessel failure ^a	139 (8.5)	80 (7.4)	59 (10.9)	0.69 (0.51-0.93)	0.013
Secondary end points ^b					
Target vessel failure without procedure-related MI	92 (5.7)	53 (4.9)	39 (7.2)	0.69 (0.49-0.97)	0.034
Cardiac death or target-vessel related MI	96 (5.9)	54 (5.0)	42 (7.7)	0.68 (0.48-0.97)	0.032
All-cause death	67 (4.1)	40 (3.8)	27 (5.0)	0.90 (0.63-1.28)	0.554
Cardiac death	33 (2.0)	17 (1.6)	16 (3.0)	0.64 (0.39-1.05)	0.079
Myocardial infarction	74 (4.5)	43 (4.0)	31 (5.7)	0.74 (0.48-1.14)	0.173
Target-vessel related MI	69 (4.2)	39 (3.6)	30 (5.5)	0.68 (0.43-1.08)	0.103
Spontaneous MI ^c	18 (1.1)	9 (0.8)	9 (1.7)	0.59 (0.26-1.33)	0.201
Procedure-related MI ^d	52 (3.2)	30 (2.7)	22 (4.0)	0.71 (0.41-1.24)	0.228
Nontarget vessel related MI	6 (0.4)	4 (0.4)	2 (0.4)	1.02 (0.30-3.45)	0.972
Repeat revascularization ^e	89 (5.6)	58 (5.4)	31 (5.8)	0.77 (0.53-1.12)	0.169
Target vessel revascularization	61 (3.8)	36 (3.4)	25 (4.7)	0.70 (0.44-1.10)	0.122
Target lesion revascularization	47 (2.9)	27 (2.5)	20 (3.8)	0.69 (0.41-1.15)	0.151
Nontarget vessel revascularization	40 (2.5)	28 (2.6)	12 (2.3)	0.75 (0.44-1.30)	0.308
Definite stent thrombosis ^f	5 (0.3)	1 (0.1)	4 (0.7)	0.13 (0.01-1.20)	0.072
Contrast induced nephropathy ^g	40 (2.4)	26 (2.4)	14 (2.6)	0.97 (0.49-1.92)	0.926
Bleeding, nontraumatic spontaneous					
BARC type 2, 3, or 5	36 (2.2)	22 (2.0)	14 (2.6)	1.03 (0.57-1.87)	0.915
BARC type 3 or 5	19 (1.2)	13 (1.2)	6 (1.1)	1.48 (0.63-3.48)	0.365
TIMI major	8 (0.5)	7 (0.6)	1 (0.2)	6.73 (0.88-51.221)	0.066
2-years to end of follow-up					
Primary end point					
Target vessel failure ^a	48 (3.8)	29 (3.4)	19 (4.5)	0.72 (0.40-1.29)	0.267
Secondary end points ^b					
Target vessel failure without procedure-related MI	48 (3.7)	29 (3.3)	19 (4.4)	0.72 (0.40-1.29)	0.270
Cardiac death or target-vessel related MI	38 (2.9)	24 (2.8)	14 (3.3)	0.80 (0.41-0.55)	0.500
All-cause death	66 (4.7)	46 (4.9)	20 (4.4)	1.09 (0.64-1.85)	0.749
Cardiac death	32 (2.5)	20 (2.3)	12 (2.8)	0.77 (0.38-1.59)	0.488
Myocardial infarction	12 (0.8)	8 (0.8)	4 (0.8)	0.93 (0.28-3.10)	0.904
Target-vessel related MI	6 (0.4)	4 (0.4)	2 (0.4)	0.96 (0.18-5.28)	0.967
Spontaneous MI ^c	6 (0.4)	4 (0.4)	2 (0.4)	0.99 (0.18-5.39)	0.987
Procedure-related MI ^d	0	0	0	-	-
Nontarget vessel related MI	6 (0.4)	4 (0.4)	2 (0.4)	0.89 (0.16-4.90)	0.897
Repeat revascularization ^e	32 (2.3)	15 (1.6)	17 (3.6)	0.42 (0.21-0.84)	0.015
Target vessel revascularization	16 (1.2)	9 (1.0)	7 (1.5)	0.64 (0.24-1.72)	0.374
Target lesion revascularization	13 (0.9)	8 (0.8)	5 (1.1)	0.80 (0.26-2.44)	0.692
Nontarget vessel revascularization	17 (1.2)	6 (0.6)	11 (2.3)	0.25 (0.09-0.68)	0.006
Definite stent thrombosis ^f	0	0	0	-	-
Contrast induced nephropathy ^g	0	0	0	-	-
Bleeding, nontraumatic spontaneous					
BARC type 2, 3, or 5	15 (1.1)	13 (1.5)	2 (0.4)	3.01 (0.68-13.37)	0.147
BARC type 3 or 5	10 (0.7)	9 (1.0)	1 (0.2)	3.99 (0.50-31.57)	0.190
TIMI major	7 (0.5)	7 (0.7%)	0 (0.0)	-	-

Values are n (%) unless otherwise indicated. The database for the analysis presented here was locked on May 10, 2025. Clinical end points were evaluated in the intention-to-treat population during the overall study period (i.e., beginning from time of randomization to the day of the first occurrence of a primary end-point event, the day of the last office or phone visit, or the day of death during follow-up). Percentages are Kaplan-Meier estimates at 2-year or at 6-year. All the models were adjusted for the clinical presentation and participating centers (stratification factor s). ^aPrimary end point is target vessel failure, which is a composite of cardiac death, target vessel MI, target vessel revascularization. ^bThe individual end points listed are the first occurrence of that event. ^cSpontaneous myocardial infarction is defined using the 3rd universal definition of myocardial infarction. ^dProcedure-related myocardial infarction is defined as the Society for Cardiovascular Angiography and Interventions (SCAI) definition. ^eRepeat revascularization includes all first clinically indicated elective, urgent, or emergent revascularization procedures that were not planned during the index hospitalization during the overall study period. ^fDefinite stent thrombosis is defined as Academic Research Consortium criteria. ^gContrast-induced nephropathy is defined as an increase in serum creatinine of ≥ 0.5 mg/dL or $\geq 25\%$ from baseline within 48-72 hours after contrast agent exposure. Event rate is presented as proportion among groups.

MI = myocardial infarction; PCI = percutaneous coronary intervention.

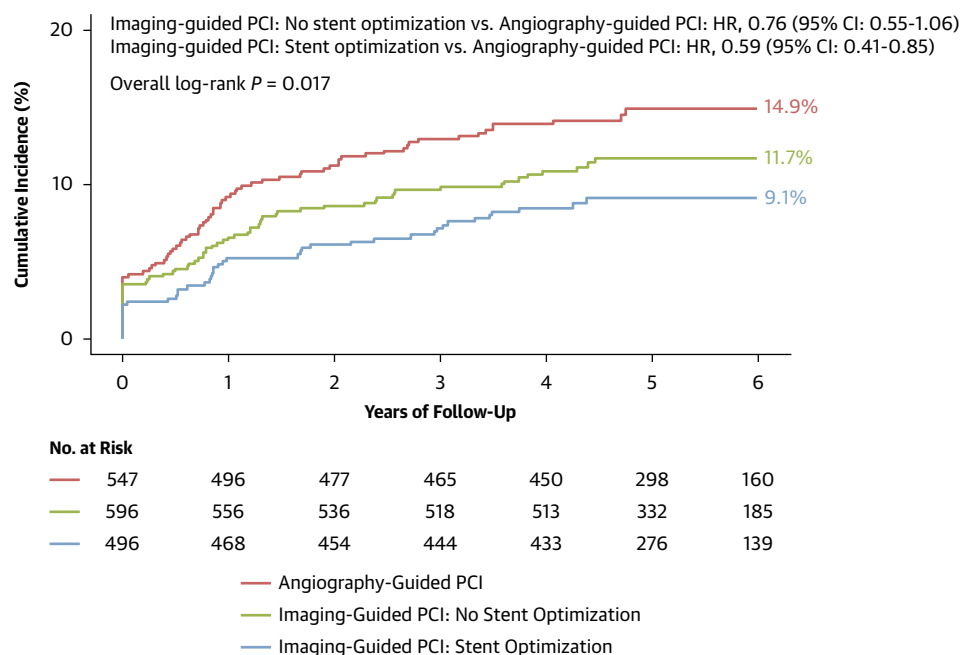
FIGURE 4 Prespecified Subgroup Analysis of Primary End Point

The hazard ratio for the primary end point was calculated in the subgroups by the choice of intravascular imaging device (IVUS or OCT), type of complex coronary lesion, initial presentation (stable ischemic heart disease or acute coronary syndrome), age (<65 years or $\geq$ 65 years), sex (male or female), diabetes mellitus, chronic kidney disease, and left ventricular ejection fraction (<50% or $\geq$ 50%). *P* values are interaction *P* values. IVUS = intravascular ultrasound; OCT = optical coherence tomography; PCI = percutaneous coronary intervention.

more prominent benefit from intravascular imaging-guided PCI than those with acute coronary syndrome. There was no significant interaction between the number of complex lesion features (interaction $P = 0.353$), or center volume (interaction $P = 0.203$) (Supplemental Figures 2 and 3). Results of the original analysis were consistent with the primary analysis results (Supplemental Table 5). In an exploratory analysis, the incidence of the primary end point was 9.1% in patients with optimized results, 11.7% in those without optimized results, and 14.9% in the angiography-guided PCI group, respectively (Figure 5).

DISCUSSION

This extended follow-up of the RENOVATE-COMPLEX-PCI demonstrates that intravascular imaging-guided PCI for complex coronary artery lesions is associated with a sustained reduction in the risk of a composite of cardiac death, target vessel-related myocardial infarction, or target vessel revascularization compared with angiography-guided PCI over a median of 5.3 years. The risk reduction was particularly evident for hard clinical end points, including cardiac death and target vessel-related myocardial infarction, and the benefit was most

FIGURE 5 Exploratory Analysis According to Treatment Group and Intravascular Imaging-Guided Optimization Results

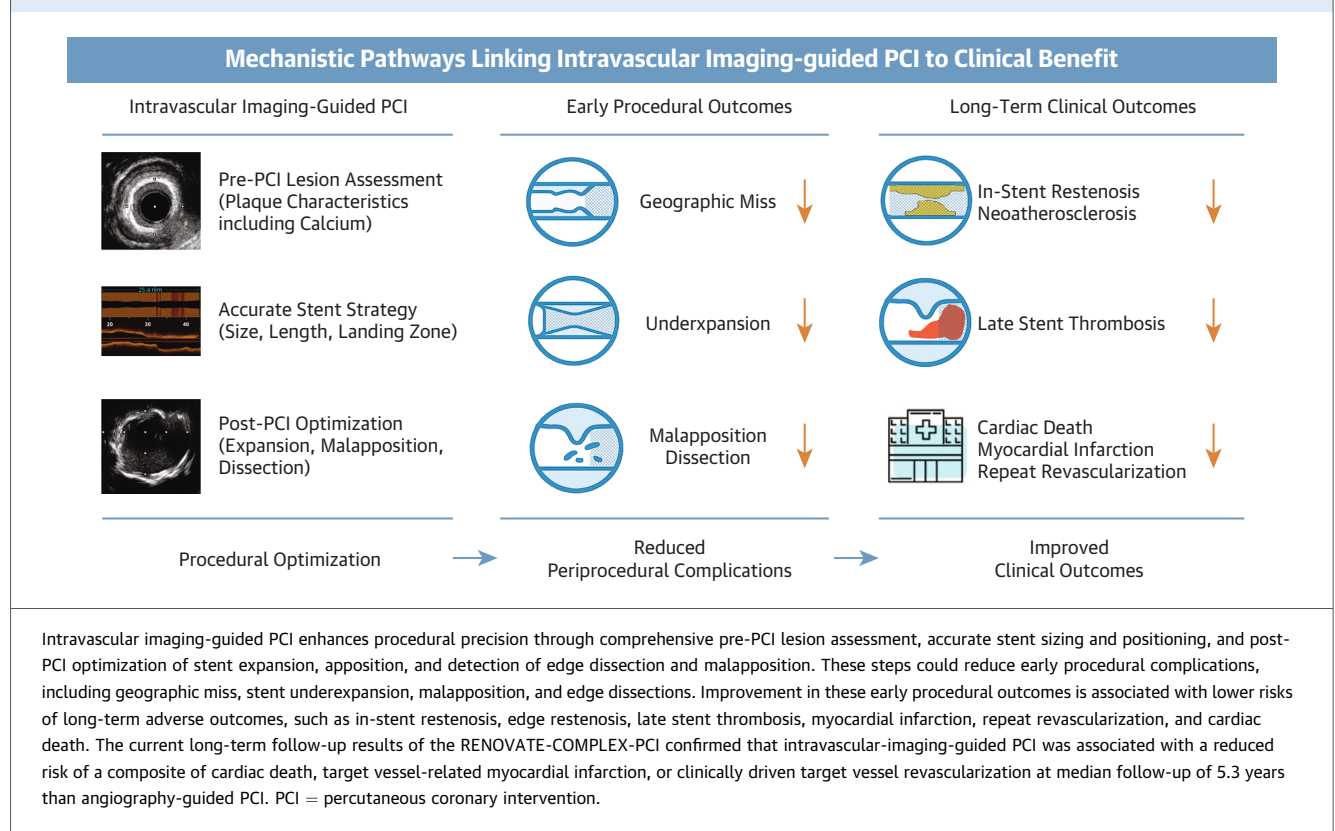
Comparison of the primary end point according to randomization group and intravascular imaging-guided optimization is shown. The primary end point was a composite of cardiac death, target vessel-related myocardial infarction, or clinically driven target vessel revascularization during the overall study period (ie, beginning from the time of randomization to the day of the first occurrence of a primary end-point event, the day of the last office or phone visit, or the day of death during follow-up). This analysis was exploratory. There was no prespecified plan to adjust for multiple testing. Outcomes are reported with 95% CIs, without P values. The 95% CIs are not adjusted for multiple testing and should not be used to infer definitive treatment effects. PCI = percutaneous coronary intervention.

pronounced in the early phase, persisting without evidence of a late catch-up phenomenon.

After publication of the initial report of the RENOVATE-COMPLEX-PCI, multiple randomized trials showed a reduced risk of clinical events following IVUS- or OCT-guided PCI compared with angiography-guided PCI,⁴⁻⁸ and updated study-level meta-analyses consistently showed superior clinical outcomes following intravascular imaging-guided PCI than angiography-guided PCI.^{9,10} Although previous practice guidelines recommended the use of intravascular imaging for procedural optimization as Class IIa (Level of Evidence: B), the most recent guidelines revised their recommendation as Class I (Level of Evidence: A) in patients with complex coronary artery lesions to optimize stent implantation, based on the recent evidence.^{11,12} However, the recent trials evaluated the clinical outcomes following intravascular imaging-guided PCI within 2 years from randomization.⁴⁻⁸ Currently, there is no available evidence regarding long-term clinical outcomes after IVUS or OCT-guided PCI beyond 2 years,

and it is unclear whether an initial strategy of intravascular imaging-guided PCI would provide better long-term clinical outcomes than an initial strategy of angiography-guided PCI. Notably, previous studies in other fields have shown that early benefits of a given revascularization strategy may not persist during extended follow-up—for example, in the FAME trial, the advantage of FFR-guided PCI over angiography-guided PCI observed at 1 year was not maintained at 5 years.¹⁴ Therefore, the main purpose of our study was to determine whether intravascular imaging-guided PCI would reduce the long-term risk of target vessel failure for complex coronary lesions compared to angiography-guided PCI beyond 2 years from index procedure.

There are several unique features of the current extended follow-up from the RENOVATE-COMPLEX-PCI that differ from previous studies comparing intravascular imaging-guided PCI and angiography-guided PCI. First, the current trial provides the longest follow-up to date among recent randomized trials comparing intravascular

FIGURE 6 Mechanistic Pathways Linking Intravascular Imaging-Guided PCI to Early and Long-Term Clinical Benefit in Complex PCI

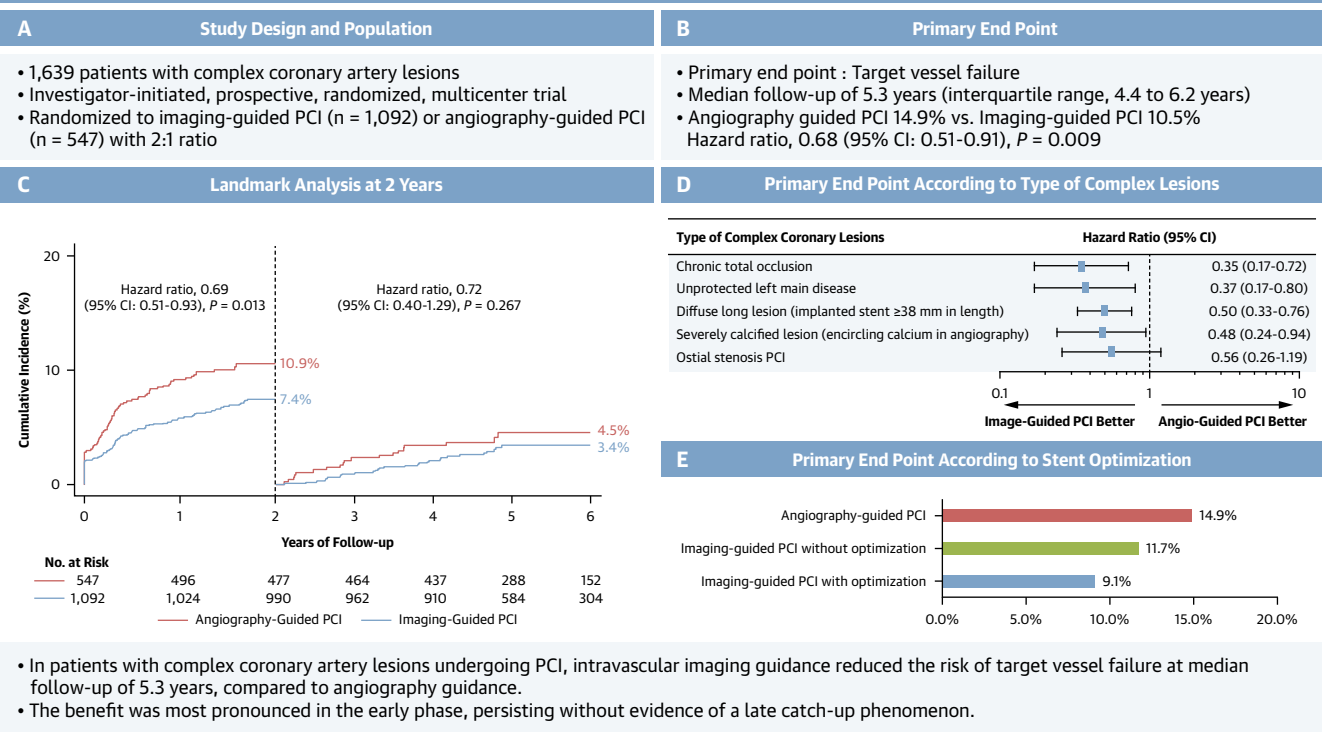
imaging- and angiography-guided PCI.⁵⁻⁸ Furthermore, all patients except 1 in the intravascular imaging-guided PCI group completed follow-up until 4 years after the last patient enrollment. This exceptionally high completeness of follow-up of trial population supports the robustness of the current results regarding long-term prognosis after intravascular imaging-guided PCI. Over a median follow-up of 5.3 years, an initial strategy of intravascular imaging-guided PCI consistently showed superior clinical outcomes than angiography-guided PCI. Regarding the effect of intravascular imaging-guided PCI, our landmark analysis presented the earlier prognostic benefit of intravascular imaging-guided PCI was maintained through later periods of follow-up without late catch-up phenomenon. Therefore, our trial supports the use of intravascular imaging-guided PCI to improve long-term clinical outcomes of patients with complex coronary artery lesions. Second, the current trial enrolled a broad spectrum of patients with complex coronary artery lesions and evaluated both of currently available intravascular imaging devices.

Prior trials evaluated more limited patient population such as patients with complex bifurcation disease,⁵ acute coronary syndrome,⁴⁻⁸ inclusion of patients with noncomplex coronary artery disease,⁶ or evaluated only of IVUS- or OCT-guided PCI.⁵⁻⁸ Therefore, the current trial would provide more generalizable results than previous evidence.

Third, the RENOVATE-COMPLEX-PCI trial firstly demonstrated that intravascular imaging-guided PCI was associated with reduction of hard clinical end points such as cardiac death or target vessel-related myocardial infarction. In our trial, the reduced risk of target vessel failure was largely due to a reduced risk of hard clinical end point including 32% lower risk of cardiac death or target vessel-related myocardial infarction, and 83% lower risk of definite stent thrombosis. Previous studies with smaller sample size or limited follow-up period showed the clinical benefit of intravascular imaging-guided PCI was mostly due to lower risk of repeat revascularization in the stented segment rather than cardiac death, myocardial infarction, or stent thrombosis.¹⁵⁻¹⁸ The recent IVUS-ACS and OCCUPI trials also observed

CENTRAL ILLUSTRATION Intravascular Imaging-Guided Versus Angiography-Guided Complex PCI: 5-Year Outcomes From the RENOVATE-COMPLEX-PCI

Five-Year Outcomes From the RENOVATE-COMPLEX-PCI



Lee JM, et al. JACC. 2026;■(■):■-■.

This extended follow-up of the RENOVATE-COMPLEX-PCI demonstrates that intravascular imaging-guided PCI for complex coronary artery lesions is associated with a sustained reduction in the risk of a composite of cardiac death, target vessel-related myocardial infarction, or target vessel revascularization compared with angiography-guided PCI over a median of 5.3 years. The risk reduction was particularly evident for hard clinical end points, including cardiac death and target vessel-related myocardial infarction, and the benefit was most pronounced in the early phase, persisting without evidence of a late catch-up phenomenon. PCI = percutaneous coronary intervention.

that intravascular imaging-guided PCI had a reduced risk of target vessel-related myocardial infarction as well as target vessel revascularization, but not in the risk of cardiac death.^{7,8} These differences in clinical event profiles might be explained by our exclusive enrollment of patients with complex coronary artery lesions, larger sample size, and choice of intravascular imaging modality according to patient and lesion characteristics. Indeed, a post-hoc analysis of the RENOVATE-COMPLEX-PCI demonstrated that the prognostic benefit of intravascular imaging-guided PCI became more evident as the number of complex lesion characteristics increased.¹⁹ Post-hoc analysis of the ILUMIEN IV trial with inclusion of patients with complex lesions also showed similar results.²⁰ Regarding the possible mechanisms of improved

prognosis following intravascular imaging-guided PCI, numerically lower rates of edge restenosis in the intravascular imaging-guided PCI group also suggest the appropriate selection for stent landing zone by intravascular imaging influence the future events coming from plaques left at the edges (Figure 6).

STUDY LIMITATIONS. First, the trial protocol for extended follow-up was not pre-specified; however, the integrity of the data is strengthened by complete follow-up, independent blinded adjudication of clinical events, and minimal crossover between study arms (27 of 1,639 patients, 1.6%). Second, more than one-half of the trial population was enrolled at a single center and the trial included only East Asian patients which might limit the generalizability of the

study (Supplemental Table 6). Third, the current trial did not systematically collect the information such as quantitative coronary angiographic analysis for nontarget lesions which were not treated at index procedure. Given the small number of new nontarget vessel MI (4 events) and the absence of difference in nontarget vessel repeat revascularization between the 2 groups, the contribution of nontarget vessel lesions (the lesions that were not treated at the index procedure) appears minimal. Fourth, the trial was open-label trial and there was no additional optimization protocol except angiographic assessment in the angiography-guided PCI group. However, we minimized the risk for any potential bias by using an end point analysis with precisely defined criteria, angiographic and imaging analyses were done at the core laboratories, and clinical events were adjudicated by a committee that was blinded to group assignment. Fifth, although intravascular imaging findings might have influenced medical treatment during follow-up, guideline-directed medical therapy was strongly recommended in both groups. Discharge medications were comparable between the two groups; however, detailed data on medication use during follow-up were not systematically collected, which precludes further analysis of potential treatment effects.

CONCLUSIONS

In patients with complex coronary artery lesions undergoing PCI, intravascular-imaging-guided PCI was associated with a reduced risk of a composite of

cardiac death, target vessel-related myocardial infarction, or clinically driven target vessel revascularization at median follow-up of 5.3 years than angiography-guided PCI (Central Illustration).

DATA AVAILABILITY Anonymized patient level data will be made available after discussion with the Executive Committee members of the trial for reasonable requests. Consent was not obtained for data sharing but the presented data are anonymized and risk of identification is minimal.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

This trial is investigator-initiated with grant support from Abbott Vascular and Boston Scientific. Other than providing financial support, the sponsors were not involved with the protocol development or study process, including site selection, study management, data collection, and analysis of the results. Dr Joo Myung Lee received an institutional research grant from Abbott Vascular, Boston Scientific, Philips Volcano, Terumo Corporation, Zoll Medical, and Donga-ST. Prof Joo-Yong Hahn received an institutional research grant from National Evidence-based Healthcare Collaborating Agency, Ministry of Health & Welfare, Korea, Abbott Vascular, Biosensors, Boston Scientific, Daiichi Sankyo, Donga-ST, Hanmi Pharmaceutical, and Medtronic Inc. Prof Hyeon-Cheol Gwon received an institutional research grant from Boston Scientific, Genoss, and Medtronic Inc. All other authors declare that there are no competing interests to declare.

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KEY WORDS coronary artery disease, intravascular imaging, intravascular ultrasound, optical coherence tomography, percutaneous coronary intervention, randomized controlled trial

APPENDIX For supplemental methods, figures, tables, and references as well as a list of investigators and collaborators, please see the online version of this paper.